Economics
Higher level
Paper 3 – Mark Scheme

SME Predicted Paper – Hard

General marking guidance:

- Award marks for correct final answers AND for correct working. Accept alternative valid methods.
- Numerical answers should be given exactly or to 2 decimal places unless otherwise specified.
- For explanations, award marks for each distinct correct point up to the maximum.
- Reward fully and correctly labelled diagrams. Unlabelled or inaccurate diagrams receive partial credit only.
- Accept valid alternative economic reasoning where the logic is consistent.
- For read-off-the-graph answers, accept any value within a reasonable band around the intended figure.

Question 1 [30 marks]

Q1 (a) (i) — Define concentration ratio [2]

1 mark: The percentage of total market output/sales/revenue accounted for by the largest X firms in the industry.

1 mark: Used as an indicator of market power / degree of competition (a 5-firm concentration ratio of 72% indicates a tight oligopoly).

Q1 (ii) — Firm B's profit-maximising output [1]

Q ≈ 66 million units (where MC = MR). Accept 62–70 million units read from Figure 1.

Q1 (iii) — Short-run economic profit/loss at (a)(ii) [2]

1 mark: Correct method: Profit = (P − AC) × Q, where P is read up from the profit-max quantity to the AR curve.

1 mark: At Q ≈ 66m, P ≈ $480, AC ≈ $332 (read from Figure 1). Profit ≈ ($480 − $332) × 66,000,000 ≈ $9.77 billion (accept $9.0–$10.5 bn with valid working). Profit is positive.

Q1 (iv) — Price charged + why abnormal profit persists [2]

1 mark: Price ≈ $480 (read up from Q = 66m to AR curve).

1 mark: Abnormal profit persists because oligopolistic markets have high barriers to entry (capital requirements, brand loyalty, patents/IP, network effects, economies of scale); new firms cannot easily enter to compete away the profit.

Q1 (v) — Kinked demand curve and price rigidity [4]

1 mark: Correctly drawn kinked AR at the prevailing price (elastic above, inelastic below).

1 mark: MR curve with a vertical discontinuity at the kink quantity.

1 mark: MC curve passing through the vertical section of the MR.

1 mark: Explanation — small shifts of MC within the discontinuity leave profit-max P and Q unchanged, so prices remain rigid despite changing conditions.

Q1 (vi) — One characteristic of common pool resources [1]

1 mark: Any one valid characteristic (rivalrous but non-excludable; vulnerable to over-use / tragedy of the commons; no clearly defined property rights).

Q1 (vii) — Total CO_2e from 66m units [2]

1 mark: Correct method: total emissions = units × per-unit emissions.

1 mark: 66,000,000 × 55 kg = 3,630,000,000 kg = 3.63 million tonnes of CO_2e.

Q1 (viii) — Externalities diagram and negative externalities [4]

1 mark: Correctly labelled externalities diagram (P, Q axes; MPC; MSC; MPB = MSB).

1 mark: Private equilibrium Qm identified; social optimum Q* identified; Qm > Q*.

1 mark: Vertical distance MSC − MPC (external cost) shown — linked to rare-earth extraction, e-waste and 55 kg CO_2e per unit.

1 mark: Welfare-loss triangle between MSC and MPC from Q* to Qm shaded and identified.

Q1 (ix) — Total indirect tax revenue [2]

1 mark: Correct method: tax revenue = tax per unit × units produced.

1 mark: $40 × 66,000,000 = $2,640,000,000 = $2.64 billion.

Q1 (b) — Policy to correct smartphone-market failures [10]

Level descriptors:

0 marks — No appropriate response.

1–3 marks (Level 1) — Little understanding; little use of terminology; no/incorrect links to data; policy identified but not meaningfully explained or evaluated.

4–6 marks (Level 2) — Some understanding; some relevant economic theory; some reference to data; policy explained but evaluation is limited and may lack balance.

7–8 marks (Level 3) — Clear understanding; accurate use of terminology; good use of data; policy well explained and reasonably evaluated; some balance.

9–10 marks (Level 4) — Thorough understanding; strong use of terminology, theory and diagrams; effective integration of data; policy fully explained, evaluated in balance (costs vs benefits, short- vs long-run, stakeholders), with a supported conclusion.

Indicative policies: Pigouvian tax on smartphone production/sales; extended-producer-responsibility (EPR) and 'right-to-repair' legislation; mandatory recycling standards; targeted subsidies for refurbished devices; tradable emissions permits for upstream extractors; competition-policy actions against anti-competitive behaviour.

Evaluation: PED of smartphones (relatively inelastic — tax may be passed to consumers); oligopolists' market power; international coordination; regressive impact of consumption taxes; innovation incentives (patents); behavioural nudges vs. hard regulation.

Question 2 [30 marks]

Q2 (a) (i) — Real GDP 2022 [2]

1 mark: Correct method: Real GDP = (Nominal GDP / GDP Deflator) × 100.

1 mark: (7,245 / 118.5) × 100 = R6,113.92 billion.

Q2 (ii) — Inflation rate 2021–2022 [2]

1 mark: Correct method: inflation = $(CPI_2 - CPI_1) / CPI_1 \times 100$.

1 mark: (116.4 − 108.0) / 108.0 × 100 = 7.78%.

Q2 (iii) — Define cyclical unemployment [2]

1 mark: Unemployment caused by a fall in aggregate demand / during the downturn of the business cycle.

1 mark: Workers lose jobs because firms reduce output and therefore demand for labour.

Q2 (iv) — One limitation of official unemployment rate [2]

1 mark: Identify a valid limitation (e.g. excludes discouraged workers; ignores underemployment; excludes informal sector; definitions differ across countries).

1 mark: Brief explanation.

Q2 (v) — AD/AS with rising interest rate [4]

1 mark: Correctly labelled AD/AS diagram.

1 mark: Higher interest rates → reduced consumption (higher cost of borrowing) and reduced investment (higher cost of capital) → AD shifts leftward.

1 mark: New equilibrium: lower price level (or reduced inflation rate).

1 mark: New equilibrium: lower real GDP / negative output gap; unemployment risk rises.

Q2 (vi) — Current account balance 2022 [2]

1 mark: Correct method: CA = (Xg − Mg) + (Xs − Ms) + NPI + NSI.

1 mark: (123.4 − 117.8) + (14.1 − 16.2) + (−12.8) + (−2.3) = 5.6 − 2.1 − 12.8 − 2.3 = −$11.6 billion.

Q2 (vii) — Two implications of a rising current account deficit [4]

2 × 2 marks (identification + explanation). Accept any two of:

- Downward pressure on the rand (depreciation).
- Must be financed by capital inflows / foreign borrowing → rising external debt.
- Weak international competitiveness / deindustrialisation risk.
- Risk of capital flight if investor confidence falls.
- If imports are capital goods, may support long-run growth.

Q2 (viii) — Infrastructure as a barrier to growth [2]

1 mark: Identify the channel (lower productivity; higher firm costs; disrupted supply chains; reduced FDI attractiveness).

1 mark: Link to LRAS / potential output failing to shift rightward.

Q2 (b) — Policy to revitalise South African growth [10]

Level descriptors:

0 marks — No appropriate response.

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Indicative policies: public investment in electricity transmission and renewables; structural reform of state-owned enterprises (Eskom); skills/training to reduce structural unemployment; targeted R&D tax credits; infrastructure-focused public–private partnerships; trade-facilitation reforms.

Evaluation: fiscal constraint (high debt/GDP); policy implementation capacity; time lags; equity implications; effect on the rand and inflation; interaction with monetary policy.